Mechanics of Human Movement
Course Progress 90.2% (11 of 13)
Week 9
Week 10
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Pathological Gait Part III
Introduction to Assistive Devices for Mobility
Design Considerations: Prosthetic Foot
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Assessment

Week 11: Assignment 11

The due date for submitting this assignment has passed. Your last recorded submission was on 2026-09-25, 19:57 IST.

Due date on 2026-10-07, 23:59 IST

1. A person with knee flexion contracture will possibly exhibit: 1 Point

☑ Shorter strides
☐ Stamping
☑ Progressive crouch gait
☐ Circumduction

Yes, the answer is correct. Score: 1

Accepted answer is Shorter strides, Progressive crouch gait

2. During the initial contact, a person walks with foot slapping on the floor. Where is the potential muscle weakness? 1 Point

○ In ankle plantarflexors
○ In quadriceps
● In ankle dorsiflexors
○ In hamstrings

Yes, the answer is correct. Score: 1

Accepted answer is In ankle dorsiflexors

3. Talipes Equinus is a disorder caused due to possible: 1 Point

☑ Contracture of gastrocnemius
☑ Contracture of soleus
☐ Amputation
☑ Weakness in tibialis anterior

Yes, the answer is correct. Score: 1

Accepted answer is Contracture of gastrocnemius, Contracture of soleus, Weakness in tibialis anterior

4. Which of the following is NOT a function of a prosthetic foot? 1 Point

○ Attenuation of shock at heel strike
● Posterior progression of the center of pressure
○ Release of stored energy for the swing phase
○ Provide sufficient friction at the foot-ground interface

Yes, the answer is correct. Score: 1

Accepted answer is Posterior progression of the center of pressure

5. Talipes Calcaneus is a disorder caused due to: 1 Point

☑ Spasticity in tibialis anterior
☐ Weakness in hamstrings
☑ Weakness in triceps surae
☐ Spasticity in quadriceps

Yes, the answer is correct. Score: 1

Accepted answer is Spasticity in tibialis anterior, Weakness in triceps surae

6. A prosthetic foot generally consists of: 1 Point

☑ Cosmetic foam
☑ Keel
☐ Socket
☐ Pylon

Yes, the answer is correct. Score: 1

Accepted answer is Cosmetic foam, Keel

7. _______ is commonly known as Club foot deformity. 1 Point

● Talipes Calcaneus
○ Genu Valgus
○ Genu Recurvatum
○ Talipes Equinovarus

No, the answer is incorrect. Score: 0

Accepted answer is Talipes Equinovarus

8. Genu Valgum (valgus) corresponds to: 1 Point

○ Bow legged appearance
● Knock knee
○ Back knee
○ Normal knee

Yes, the answer is correct. Score: 1

Accepted answer is Knock knee

9. Which of the following are components of a below-knee modular prosthesis? 1 Point

☑ Foot
☐ Knee
☑ Socket
☐ Extension assist

Yes, the answer is correct. Score: 1

Accepted answer is Foot, Socket

10. In the below figure, the heel of a prosthetic foot is shown to be compressed. What are the possible direct functions of this event? 1 Point

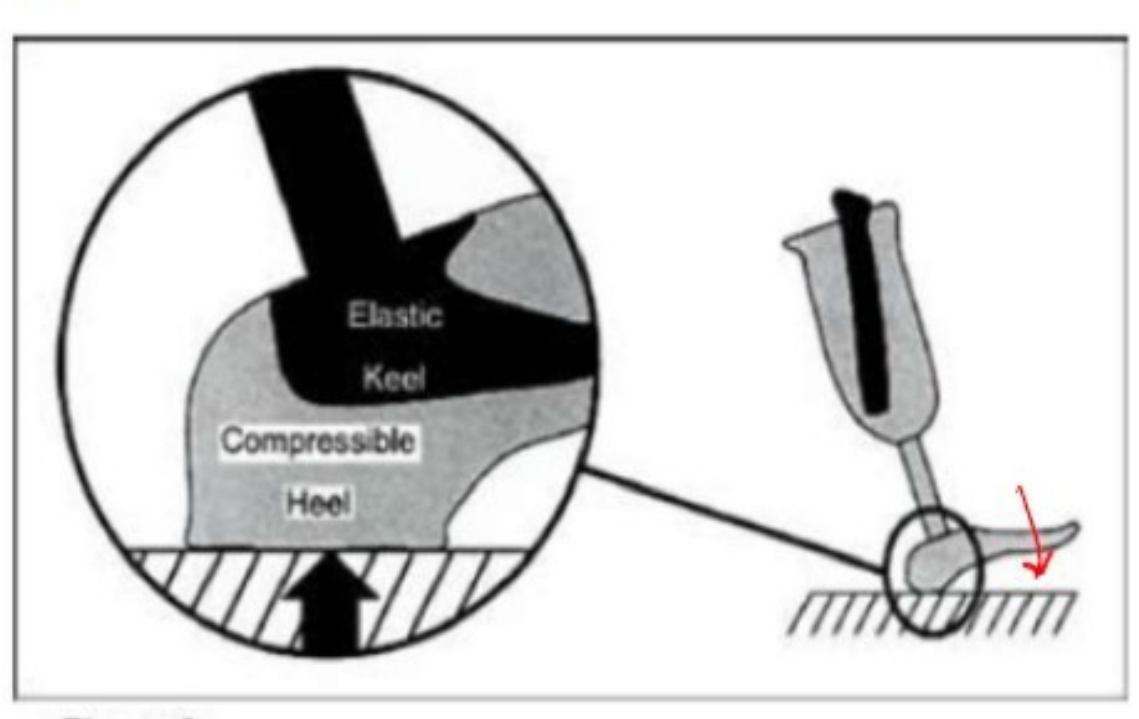

Figure 6.
Heel compression of a prosthesis during loading response.

☑ Absorption of shock at heel strike
☑ Transfer of weight onto the prosthesis
☑ Storage of energy into the prosthetic foot
☐ Assistance in push off

Yes, the answer is correct. Score: 1

Accepted answer is Absorption of shock at heel strike, Transfer of weight onto the prosthesis, Storage of energy into the prosthetic foot

11. In the below figure, the keel of a prosthetic foot is shown to be loaded during terminal stance. What are the direct functions of this event? 1 Point

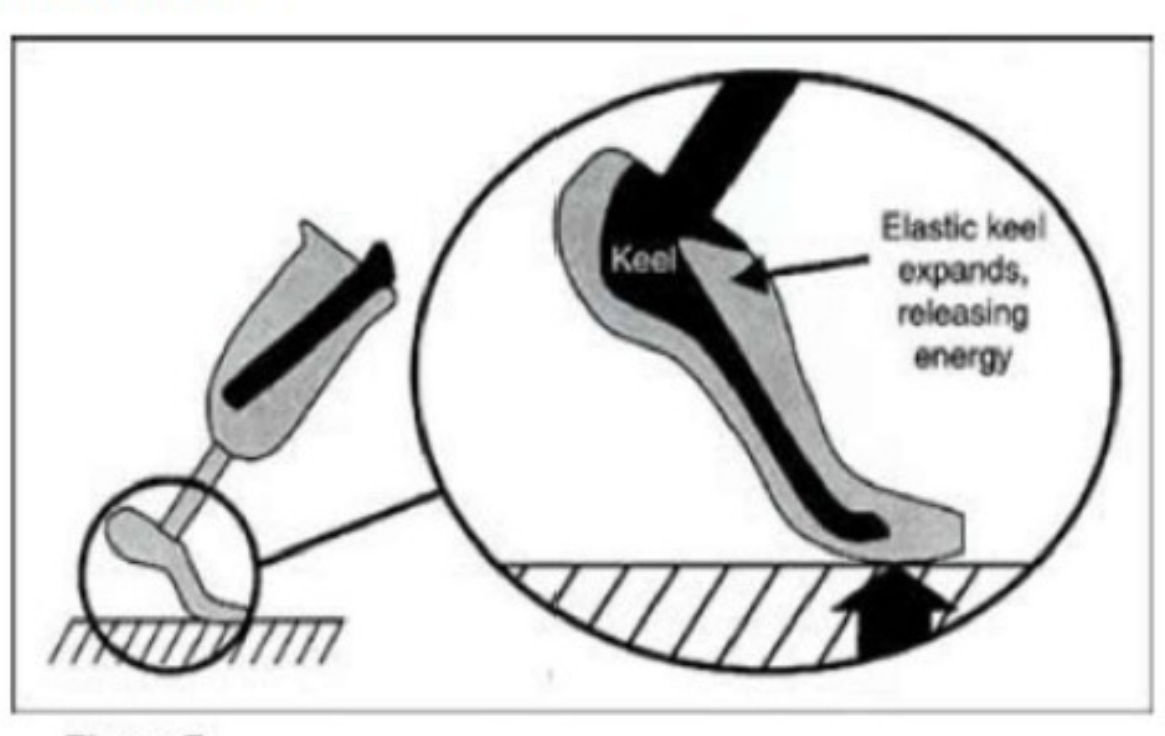

Figure 7.
Keel loading of a prosthesis during terminal stance.

☑ Storage of energy during loading
☑ Assistance in push off
☐ Attenuation of shock during loading response
☐ Control of pendular swing

Yes, the answer is correct. Score: 1

Accepted answer is Storage of energy during loading, Assistance in push off